

Generation Interconnection System Impact Study Report

For Queue Project Q17-0901

Monfort 100 kV
November 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0901. The Interconnection Customer proposes a 56.6 MW natural gas combined cycle facility interconnecting at the Monfort 100 kV substation in the Indiana area. The request was received on December 12, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 268 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,776,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Corridor / Larson St 230kV Line	Trego / Beaver 500kV L	111.7	851	8.5
Lonsdale / Hubbard 100kV	Richland / Dekalb 500kV	111.1	126	8.3
Lookingglass / Truman 230kV Line	Trego / Beaver 500kV	110.9	911	8.5
Waldo / Truman 230kV line	Beaver / Trego 500kV Line	109.1	949	8.5
Troy Energy Llc / Richland 500kV Line	Swan Valley / Trego 500kV	108.7	3,510	19.3
Lonetree 500 kV/230 kV Transformer Bank	East Methuen / Lonetree 500kV	108.3	811	6.3
Hebo / BurrisPit 230kV	Trego / Beaver 500kV	107.9	1,021	9.3
Baraga Vlg / Gray'S Flat Substation 230kV	Madeline / E Methuen 230kV	107.9	1,020	11.1
Burris Pit / James De Young 230kV	Beaver / Trego 500kV line	107.4	1,046	9.3
Sabanera 500 kV/230 kV TX	Lonetree / Sabanera 500kV	107.4	1,302	11.1
Henry County 500/230 kV XFMR	Sabanera / Loetree 500kV	106.8	894	9.5
Gray'S Flat / East Methen 230kV	East Methuen Substation / Madeline 230kV L	106.1	1,052	11.1
Madeline Substation / Baraga Vlg 230kV	E Methuen / Gray'S Flat 230kV Line	105.8	637	8.2
Lookingglass / Corridor 230kV line	Trego / Beaver 500kV	105.7	926	8.5
Minnehaha / Sta S Van Nuys 230kV L	Richland / Troy Energy Llc 500kV	105.7	1,336	6.8
Richland / Dekab 500kV	Slocum Rd / Dekalb 230kV	104.9	3,847	15.7
Waldo 500 kV/230 kV Auto	Trego / Beaver 500kV	104.6	1,089	8.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mason St 500-230 kV Auto	Beaver / Trego 500kV	104.3	1,705	9.0
Adirondack / James De Young 230kV L	James De Young / St. Lawrence 230kV	102.9	952	8.7
Beaver / Onancock Diesel Generation Facility Substation 230kV Line	Eufaula / Beaver 500kV	102.5	1,730	10.5
Troy Energy Llc / Trego 500kV Line	Swan Valley / Trego 500kV line	102.4	3,450	18.7
15Th Avenue S / Lucent 230kV L	Omni-Hackberry / Norwich 500kV	102.2	784	5.4
Carley Rd / James De Young 230kV	East Methuen Substation 500 kV/230 kV Transformer Bank	102.0	781	8.6
Galion Generating Station 500/230kV Autotransformer	Beaver / Trego 500kV	101.9	1,417	9.6
15Th Avenue S / Norwich 230kV	Omni-Hackberry / Norwich 500kV	101.9	801	5.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Corridor / Larson Street 230kV	Rebuild line with higher-capacity conductor	\$12,411,000	1% (\$73,000)
Lonsdale / Hubbard 100kV	Rebuild line with higher-capacity conductor	\$2,479,000	21% (\$526,000)
Lookingglass / Truman 230kV Line	Rebuild line with higher-capacity conductor	\$15,370,000	1% (\$91,000)
Waldo / Truman 230kV line	Reconductor line	\$10,229,000	1% (\$60,000)
Troy Energy Llc / Richland 500kV Line	Reconductor line	\$70,946,000	0% (\$336,000)
Lonetree 500/230 kV TX	Replace transformer with higher-rated unit	\$45,420,000	1% (\$546,000)
Burris Pit / Hebo 230kV	Reconductor line	\$7,536,000	1% (\$46,000)
Gray'S Flat / Baraga Vlg 230kV	Reconductor line	\$5,400,000	1% (\$31,000)
Burris Pit / James De Young 230kV L	Reconductor line	\$11,305,000	1% (\$70,000)
Sabanera 500 kV/230 kV TX	Replace transformer with higher-rated unit	\$31,020,000	1% (\$212,000)
Henry County 500/230 kV XFMR	Replace transformer with higher-rated unit	\$28,798,000	1% (\$190,000)
Gray'S Flat / East Methen 230kV	Reconductor line	\$15,005,000	1% (\$85,000)
Madeline / Baraga Vlg 230kV	Reconductor line	\$5,375,000	1% (\$30,000)
Lookingglass / Corridor 230kV line	Reconductor line	\$8,305,000	1% (\$49,000)
Minnehaha / Sta S Van Nuys 230kV L	Reconductor line	\$6,075,000	0% (\$27,000)
Richland / Dekalb 500kV	Reconductor line	\$54,086,000	0% (\$188,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Waldo 500 kV/230 kV TX	Replace transformer with higher-rated unit	\$17,054,000	1% (\$101,000)
Mason St 500-230 kV Auto	Replace transformer with higher-rated unit	\$33,732,000	1% (\$186,000)
Adirondack / James De Young 230kV L	Reconductor line	\$19,029,000	1% (\$113,000)
Beaver / Onancock Diesel Generation Facility Substation 230kV Line	Reconductor line	\$7,185,000	1% (\$43,000)
Troy Energy Llc / Trego 500kV Line	Reconductor line	\$76,198,000	0% (\$376,000)
15Th Avenue S / Lucent 230kV L	Reconductor line	\$3,192,000	1% (\$25,000)
Carley Rd Substation / James De Young 230kV line	Reconductor line	\$11,842,000	0% (\$65,000)
Galion Generating Station 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$45,039,000	1% (\$266,000)
15Th Avenue S / Norwich 230kV	Reconductor line	\$5,121,000	1% (\$41,000)
	Total allocated to project		\$3,776,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.